

Recession Forecasting - Yield Curve and ML

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Abstract

This paper evaluates a recession forecasting pipeline designed to predict U.S. National Bureau of Economic Research (NBER) recessions twelve months ahead. The framework integrates information from the Treasury yield curve, macroeconomic indicators, labor market variables, credit spreads, and explicit term premium decompositions. Principal component analysis (PCA) is used to extract interpretable yield curve factors, whose stability is examined through rolling-window diagnostics. Forecasting models are evaluated under strict temporal discipline using expanding-window cross-validation and out-of-sample threshold selection. Results show that yield curve slope and curvature factors are highly informative for financially driven recessions such as the Global Financial Crisis, while macroeconomic variables dominate performance during abrupt shock episodes such as the COVID-19 recession. The findings highlight the regime-dependent nature of recession signals and the importance of separating expectations-driven yield movements from term premium distortions.

1 Introduction

Forecasting economic recessions is a central problem in macroeconomics and financial economics, particularly when predictions must be made at horizons that are relevant for policy and risk management. The Treasury yield curve has long been recognized as one of the most reliable leading indicators of U.S. recessions. However, the interpretation of yield curve signals has become more complex in recent decades due to the expansion of unconventional monetary policy, large-scale asset purchases, and sustained compression of term premia.

This paper develops and evaluates a forecasting pipeline that predicts U.S. recessions twelve months in advance. Unlike contemporaneous classification approaches, this design imposes a genuine forecasting challenge. The analysis integrates yield curve information across maturities, principal component representations, macroeconomic indicators, labor market measures, and explicit decompositions that separate expectations-driven yield movements from term-premium-driven dynamics. Special attention is paid to avoiding information leakage and to diagnosing the stability of yield curve factors across monetary regimes.

2 The Yield Curve

The predictive power of the yield curve $y_t(\tau)$ regarding economic activity is best understood by decomposing the term structure into three latent factors: *Level*, *Slope*, and *Curvature*. Following the standard framework (e.g., Nelson and Siegel (1987)), the yield at maturity τ can be modeled as a function of these distinct components. The *Level* factor (L_t) represents the long-term equilibrium interest rate and inflation expectations, effectively shifting the entire curve vertically. The *Slope* factor (S_t), defined as the spread between long-term and short-term rates ($y_t(\tau_{long}) - y_t(\tau_{short})$), serves as the primary recessionary barometer; a negative slope (inversion) implies that investors anticipate future rate cuts due to economic contraction. Finally, the *Curvature* factor (C_t) captures the "hump" or convexity of the curve, representing the movement of medium-term yields relative to the short and long ends.

This theoretical decomposition is clearly visible in the transition from the pre-crisis to the recovery era shown in Figure 1. The 2006 average (blue line) exhibits a "high Level, zero Slope" dynamic, where short-term rates ($y_{3m} \approx 5\%$) have converged with long-term rates ($y_{30y} \approx 5\%$). The absence of Slope and Curvature here indicates a restrictive monetary environment where the term premium has eroded—a classic precursor to the 2008 recession. Conversely, the 2010 average (red line) illustrates a fundamental regime shift. While the *Level* at the long end remains elevated, the *Slope* has steepened dramatically ($S_t \gg 0$) as short-term rates approach the zero lower bound. Furthermore, the 2010 curve displays significant *Curvature* (concavity), reflecting the market's nuanced pricing of medium-term recovery risk against a backdrop of aggressive monetary stimulus.

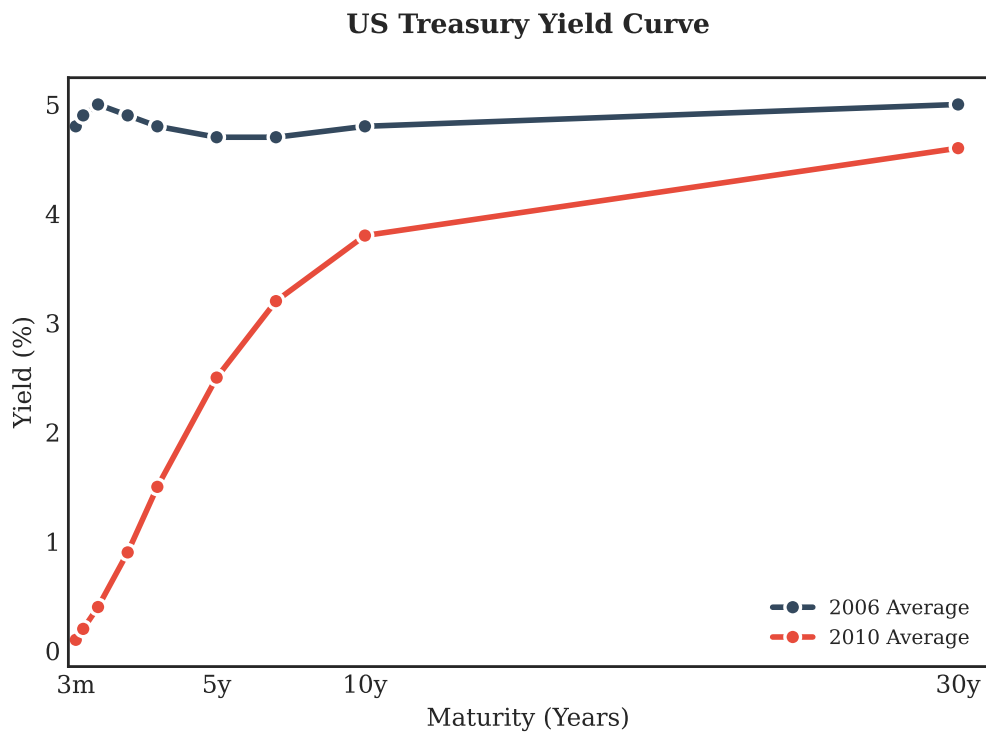


Figure 1: US Yield Curve for selected Maturities for 2006 and 2010

Source: FRED

3 Data and Target Construction

The target variable is the monthly NBER recession indicator. To ensure a true forecasting horizon, the target is shifted backward by twelve months, such that all predictors available at time t are used to forecast recession status at time $t + 12$.

The yield curve dataset consists of monthly Treasury yields spanning maturities from three months to thirty years. These yields are used both directly and as inputs to principal component analysis. Macroeconomic indicators include the unemployment rate, year-over-year CPI inflation, year-over-year industrial production growth, and consumer sentiment. Labor market conditions are proxied using year-over-year growth in nonfarm payroll employment. Financial conditions are captured through the federal funds rate, corporate credit spreads, and an estimated ten-year Treasury term premium.

To distinguish expectations-driven yield curve movements from term premium effects, the ten-year yield is decomposed into an expectations proxy and a term premium component. An adjusted slope measure is constructed by subtracting the term premium from the long rate prior to computing the slope relative to the two-year yield. This allows direct testing of whether recession signals arise from anticipated policy easing or from non-expectational forces such as quantitative easing and safe-asset demand.

4 Principal Component Analysis of the Yield Curve

4.1 Explained Variance

Figure 2 presents the scree plot of the PCA applied to standardized yields across maturities. The first principal component explains approximately 97% of total variance, while the second and third components explain roughly 2.5% and 0.1%, respectively. This indicates that the U.S. yield curve is overwhelmingly driven by a common level factor, although slope and curvature components remain economically meaningful despite their small variance shares.

Table 1: Explained variance ratio of principal components.

index	0
PC1	0.971
PC2	0.027
PC3	0.002

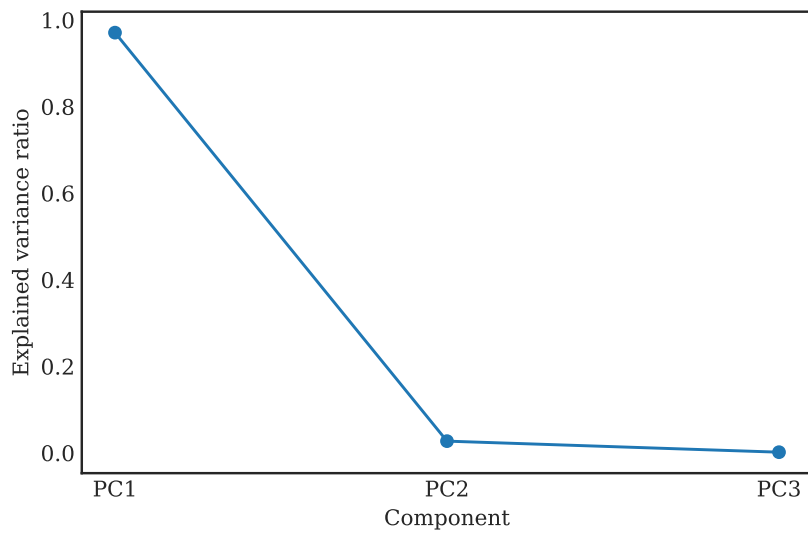


Figure 2: Explained variance ratio of yield curve principal components.

Table 1 reports the explained variance ratios numerically.

4.2 Factor Loadings

Figure 3 displays the PCA loading heatmap across maturities. The first principal component loads positively and nearly uniformly across all maturities, consistent with a level factor. The second component exhibits negative loadings at the short end and positive loadings at the long end, representing yield curve slope. The third component captures curvature, reflecting differential movements of the belly of the curve relative to short and long maturities.

Table 2: PCA loadings for yield curve components.

index	PC1	PC2	PC3
y_3m	0.329	-0.446	0.514
y_6m	0.331	-0.401	0.218
y_1y	0.334	-0.313	-0.089
y_2y	0.337	-0.140	-0.366
y_3y	0.338	-0.026	-0.421
y_5y	0.337	0.158	-0.294
y_7y	0.335	0.267	-0.132
y_10y	0.333	0.370	0.099
y_30y	0.325	0.537	0.505

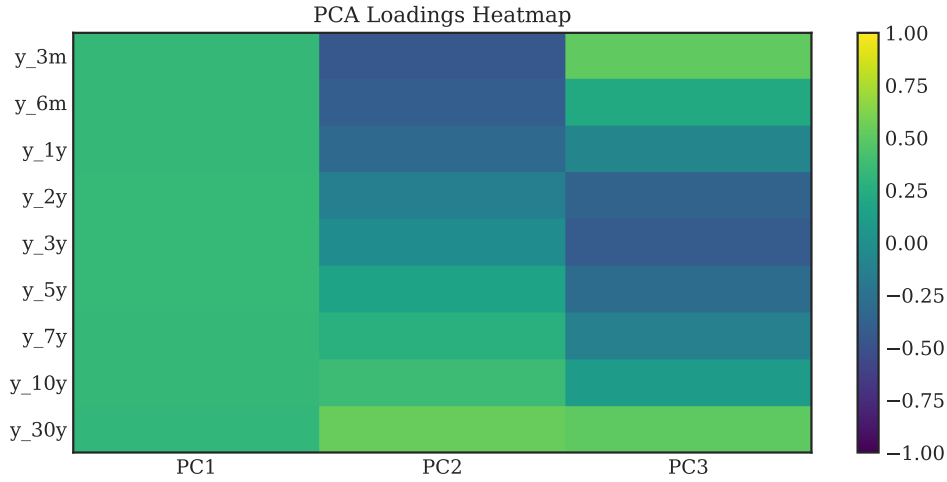


Figure 3: Heatmap of yield curve PCA loadings by maturity.

Detailed loadings are reported in Table 2.

4.3 Factor Dynamics and Recessions

Figure 4 plots the time series of the first three principal components with NBER recession shading. The level factor reflects long-run secular declines in interest rates and major monetary regime shifts. The slope factor exhibits pronounced swings around tightening

cycles and yield curve inversions preceding recessions. The curvature factor shows episodic spikes during periods of financial stress and policy intervention.

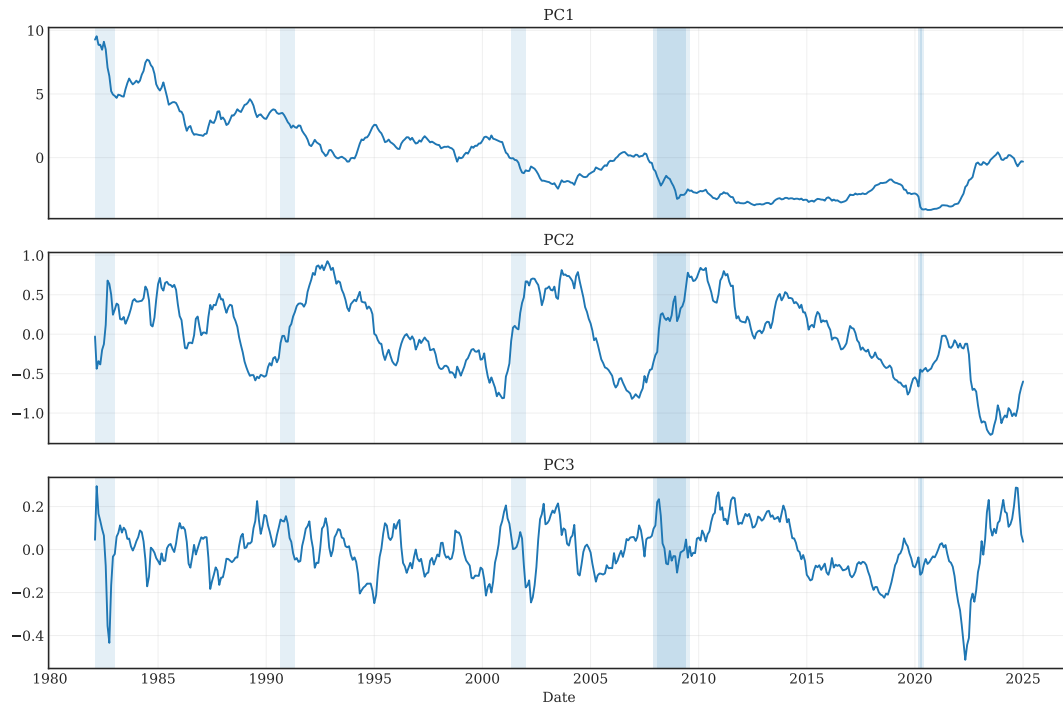


Figure 4: Time series of yield curve principal components with recession shading.

4.4 Rolling Loading Stability

Figures 5–7 present rolling-window PCA loading estimates for the first three components. While the overall factor structure is preserved, substantial drift is observed, particularly at the long end of the curve during the Global Financial Crisis and the COVID-19 period. These findings suggest that yield curve factors are regime-dependent and that long-maturity yields become structurally distorted during periods of unconventional monetary policy.

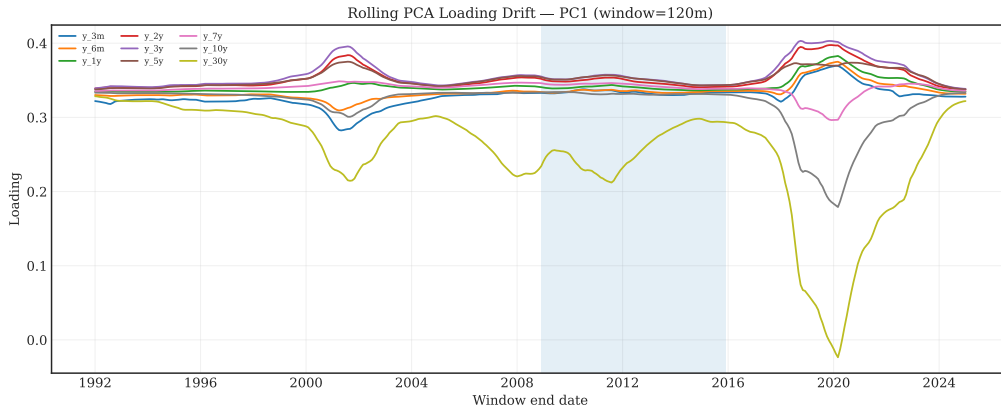


Figure 5: Rolling PCA loading drift for the level factor (PC1).

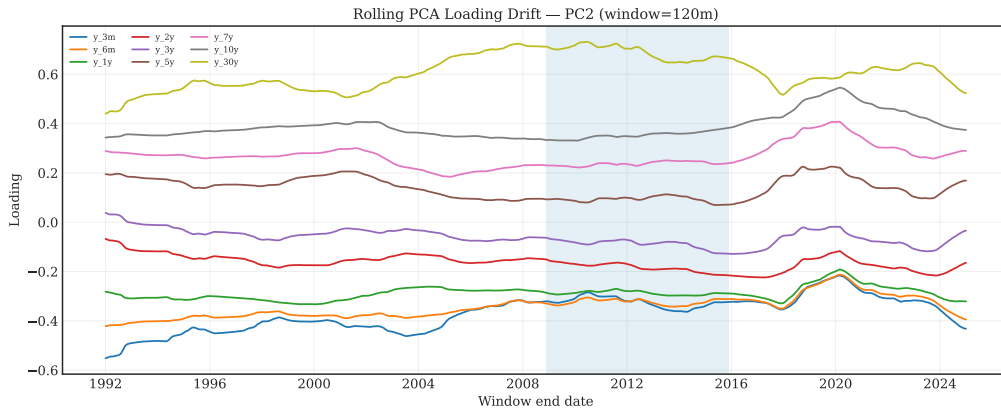


Figure 6: Rolling PCA loading drift for the slope factor (PC2).

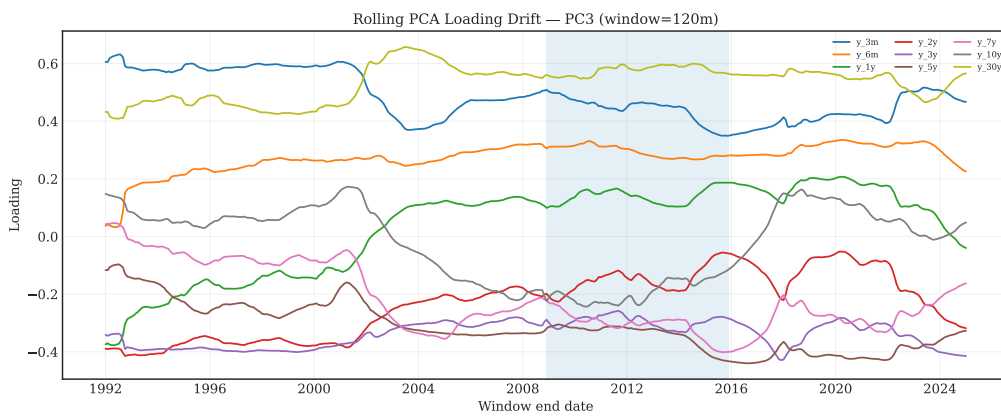


Figure 7: Rolling PCA loading drift for the curvature factor (PC3).

5 Forecasting Models and Evaluation

Models are estimated using multiple feature sets, including yield spreads, raw yields, yield PCA factors, macroeconomic variables, and hybrid combinations incorporating credit spreads and term premium proxies. Classifiers include regularized logistic regression, random forests, gradient boosting machines, and extreme gradient boosting. Thresholds are selected using out-of-fold predicted probabilities generated exclusively from the training sample via expanding-window cross-validation.

Table 3 reports holdout performance across multiple train–test splits.

6 Scenario Analysis: COVID-19 and the Global Financial Crisis

To assess regime dependence, models are evaluated separately on the COVID-19 recession (2019–2021) and the Global Financial Crisis (2007–2009). Figures 8 and 9 report F2 scores, emphasizing recall, while Figures 10 and 11 report precision–recall AUC.

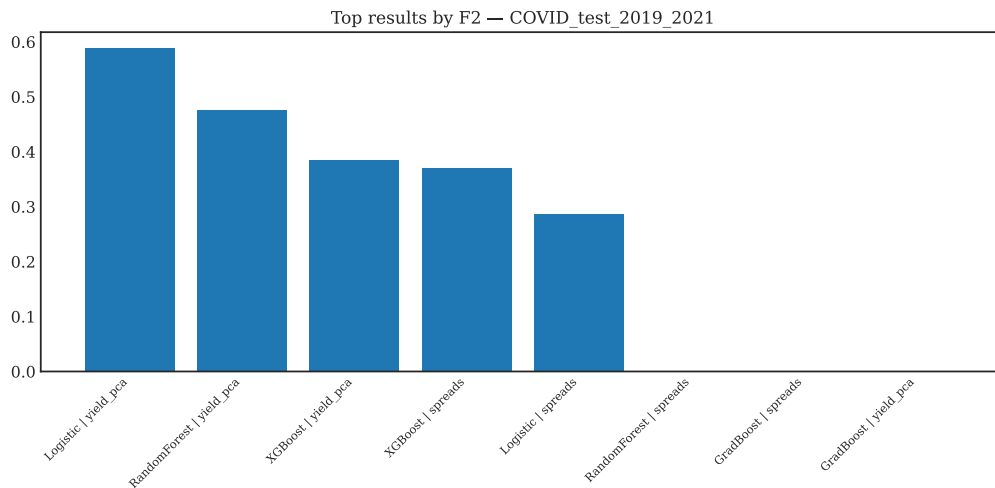


Figure 8: Top F2 scores during the COVID-19 recession.

Table 3: Holdout sensitivity results across feature sets and models.

TestFrac	FeatureSet	Model	ThrObjective	ThrOpt	ThrObjectiveValue_onTrainOOF
0.200	spreads	Logistic	balanced_acc	0.590	0.575
0.200	spreads	RandomForest	balanced_acc	0.655	0.667
0.200	spreads	GradBoost	balanced_acc	0.999	0.485
0.200	spreads	XGBoost	balanced_acc	0.096	0.624
0.200	yield_pca	Logistic	balanced_acc	0.864	0.659
0.200	yield_pca	RandomForest	balanced_acc	0.096	0.636
0.200	yield_pca	GradBoost	balanced_acc	0.909	0.589
0.200	yield_pca	XGBoost	balanced_acc	0.086	0.625
0.300	spreads	Logistic	balanced_acc	0.116	0.757
0.300	spreads	RandomForest	balanced_acc	0.096	0.680
0.300	spreads	GradBoost	balanced_acc	0.001	0.547
0.300	spreads	XGBoost	balanced_acc	0.096	0.658
0.300	yield_pca	Logistic	balanced_acc	0.146	0.702
0.300	yield_pca	RandomForest	balanced_acc	0.096	0.670
0.300	yield_pca	GradBoost	balanced_acc	0.006	0.528
0.300	yield_pca	XGBoost	balanced_acc	0.146	0.658
0.400	spreads	Logistic	balanced_acc	0.226	0.740
0.400	spreads	RandomForest	balanced_acc	0.086	0.757
0.400	spreads	GradBoost	balanced_acc	0.270	0.552
0.400	spreads	XGBoost	balanced_acc	0.086	0.745
0.400	yield_pca	Logistic	balanced_acc	0.400	0.750
0.400	yield_pca	RandomForest	balanced_acc	0.141	0.768
0.400	yield_pca	GradBoost	balanced_acc	0.001	0.665
0.400	yield_pca	XGBoost	balanced_acc	0.300	0.729

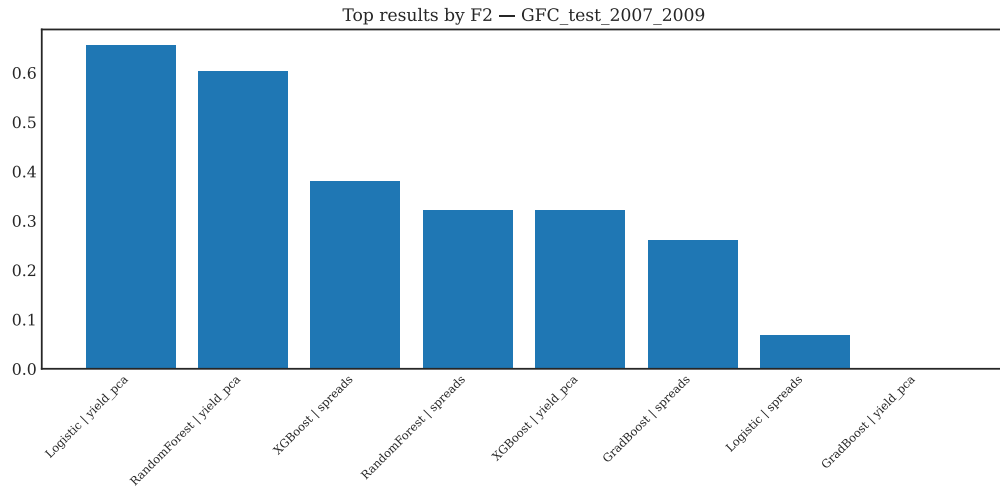


Figure 9: Top F2 scores during the Global Financial Crisis.

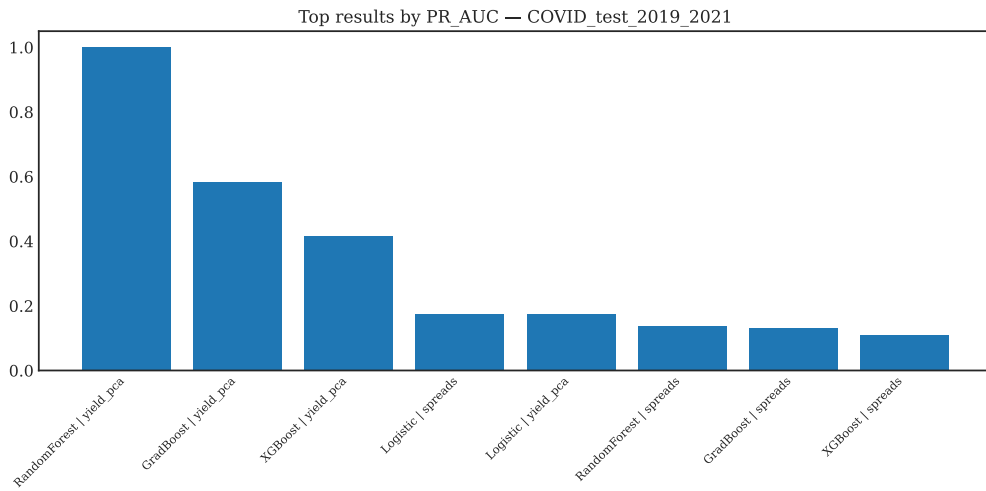


Figure 10: Top PR-AUC scores during the COVID-19 recession.

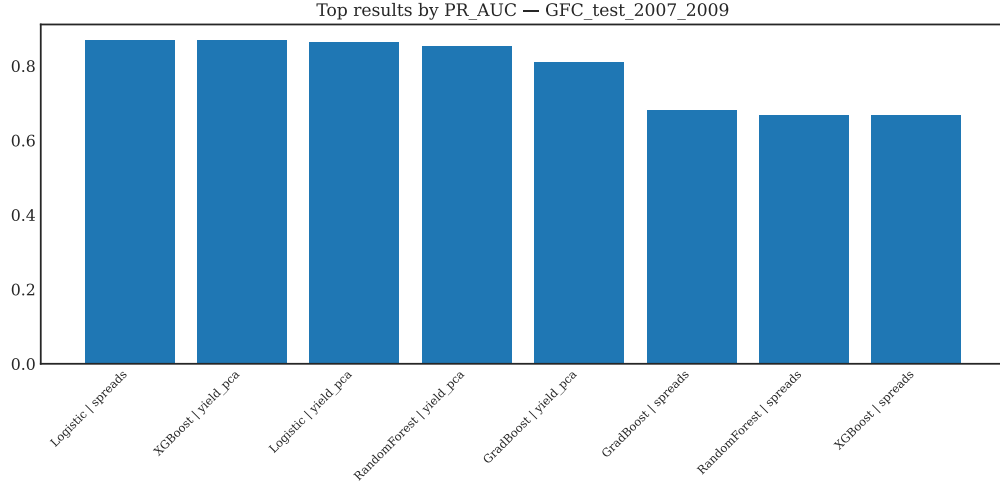


Figure 11: Top PR-AUC scores during the Global Financial Crisis.

Table 4 summarizes scenario-specific results numerically.

7 Discussion

Results demonstrate that yield curve PCA factors, particularly slope and curvature, provide strong leading information for financially driven recessions such as the Global Financial Crisis. In contrast, macroeconomic variables dominate performance during sudden shock episodes like COVID-19, where traditional yield curve signals lag the downturn. Rolling PCA diagnostics reveal that factor stability deteriorates during periods of unconventional monetary policy, underscoring the importance of term premium controls and regime-aware modeling.

8 Conclusion

This paper shows that recession forecasting performance is highly regime-dependent. Yield curve factors remain powerful predictors, but their interpretation must account for term premium distortions and structural instability. Combining PCA-based yield representations with macroeconomic indicators and disciplined threshold selection yields robust and interpretable forecasts across diverse economic environments.

Table 4: Scenario-based evaluation results (GFC vs COVID).

Scenario	TrainEnd	TestStart	TestEnd	FeatureSet	Model	Thro
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	spreads	Logistic	balan
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	spreads	RandomForest	balan
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	spreads	GradBoost	balan
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	spreads	XGBoost	balan
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	yield_pca	Logistic	balan
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	yield_pca	RandomForest	balan
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	yield_pca	GradBoost	balan
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	yield_pca	XGBoost	balan
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	spreads	Logistic	balan
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	spreads	RandomForest	balan
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	spreads	GradBoost	balan
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	spreads	XGBoost	balan
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	yield_pca	Logistic	balan
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	yield_pca	RandomForest	balan
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	yield_pca	GradBoost	balan
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	yield_pca	XGBoost	balan

References

Nelson, C. R. and Siegel, A. F. (1987). Parsimonious Modeling of Yield Curves, *The Journal of Business* **60**(4): 473–489.

URL: <https://www.jstor.org/stable/2352957>